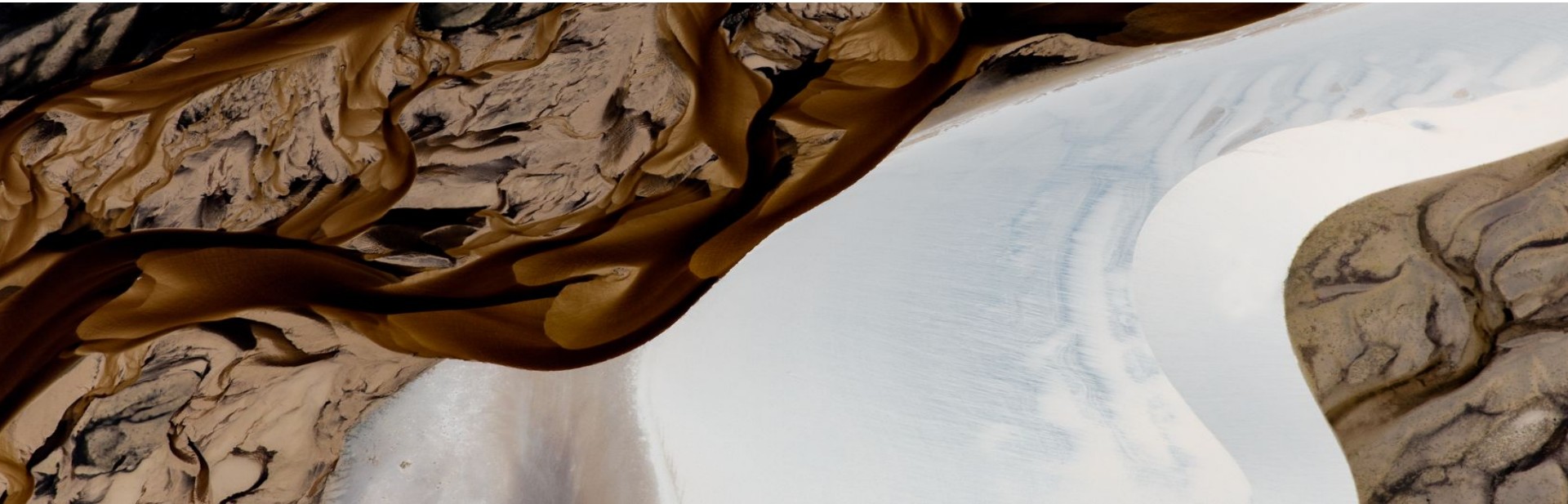
Mapping the Open Source Ecosystem for Climate Science and Sustainable Technology

Richard Littauer, Tobias Augspurger, Alex Stinson, Pierre Vogler-Finck, Andrew Nesbitt, Chris Harris
https://opensustain.tech



Why are we talking here today?

- The climate crisis is urgent, deadly, and difficult.
- Solutions will involve collaboration and ingenuity.
- Open source software offers the best possible chance of enabling a fast, effective response.
- We want to facilitate its discovery, maintenance and use.

OpenSustain.tech is the largest, curated, open database
of OSS used in climate science.

Who are we?

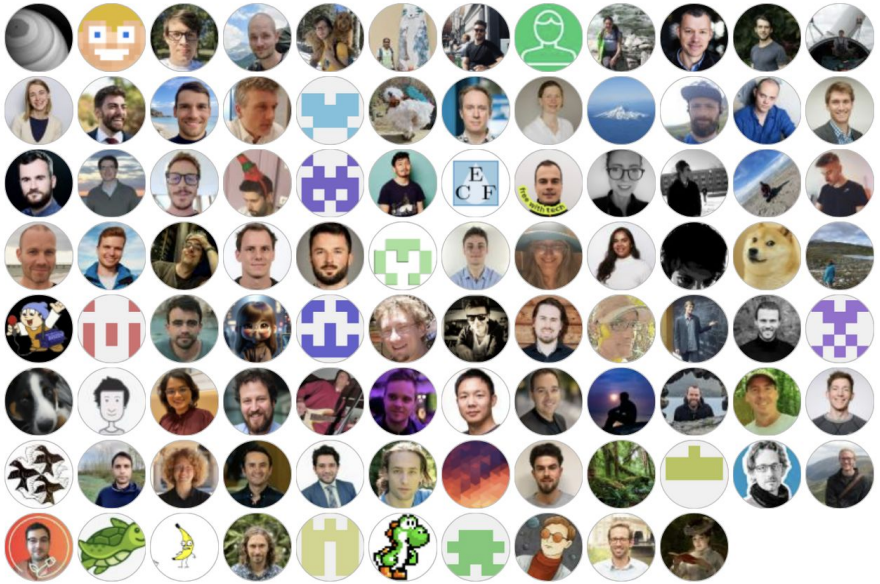
Richard Littauer

- CURIOSS, SustainOSS, PythonNZ

Alex Stinson (with apologies):

- Wikimedia Foundation, Work on Climate, Climate Drift

Contributors



The problem

Climate crisis is here.

Solving it will require work from all parts of society.

Governance alone is unlikely to resolve the crisis.

The Guardian

Int

News | Opinion | Sport | Culture | Lifestyle

World | US politics | UK | Climate crisis | Middle East | Ukraine | Environment | Science | Global development

Cop30

Rich countries have lost enthusiasm for tackling climate crisis, says Cop30 chief

Brazil's André Corrêa do Lago says countries should follow China's lead on clean energy as conference begins

- Cop30: what are the main issues?
- Net zero to NDCs: your Cop30 jargon buster



Open * can help



Open organizing, science, data, models, access, knowledge, source.

Top-down approach: funders, governmental policies.

Bottom up: individual researchers, individual projects.

OSS

- 1/3rd of all of repositories on GitHub
- 70-90% of software has open source in it (LF, 2022); 97% (Black Duck, 2025)
- Supply side worth is $4 billion; demand $9 trillion (Nagle et al.).

OSS in climate science

- No clear, predictable links
- Big Tech is disincentivized to release open models
- Government accountability is unreliable globally
- DPGs may provide some help in the future.

Open Sustain Technology

- https://opensustain.tech
- OSS-community driven project
- Started by Tobias Augspurger in 2020
- Diverse team - academics, climate scientists, OSS ecosystem workers.
- Collaborations with Linux Foundation, GitHub, and others.

Finding projects themselves is hard

- No clear internet-wide search engines for OSS
- Online marketing and AI slop causes semantic bleaching
- Polysemic terms: "cloud", "climate", "environment"
- Flood of low-quality OSS projects
- GitHub stars overinflate usage metrics
- No standardized citation format for used papers
- No standardized protocols for policy usage

List-based approach

- Prioritizes human curation
- Enables collaboration and re-use under open licensing
- Enables loose definitions of inclusion
- Uses existing systems (PRs, Markdown) in tech

Open Sustainable Technology's mission is to find, list, and share projects that preserve or analyze natural ecosystems through open technology, methods, data, intelligence, knowledge or tools.

Criteria

1. OSS license?
2. Demonstrable usage and goals that are relevant to environmental sustainability?
3. Active usage or development in the last year?
4. Use outside the project (citations, opened issues, citations or downloads)?
5. Documentation for re-use?

Sections

Human systems:

- Renewable Energy
- Energy Storage
- Energy Systems
- Consumption
- Emissions
- Industrial Ecology
- Sustainable Development

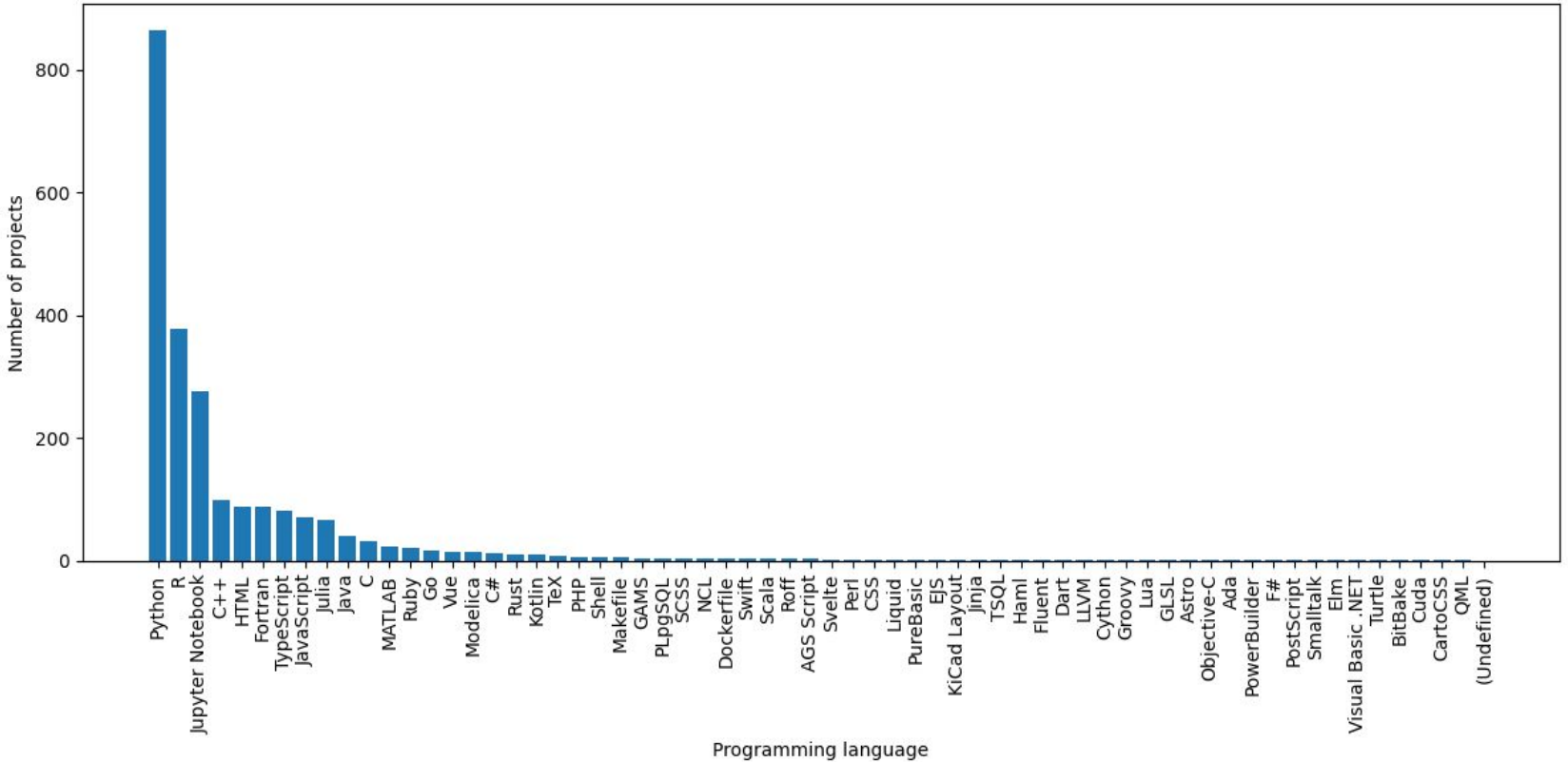
Natural resources:

- Biosphere
- Cryosphere
- Hydrosphere
- Atmosphere
- Climate Change
- Natural Resources

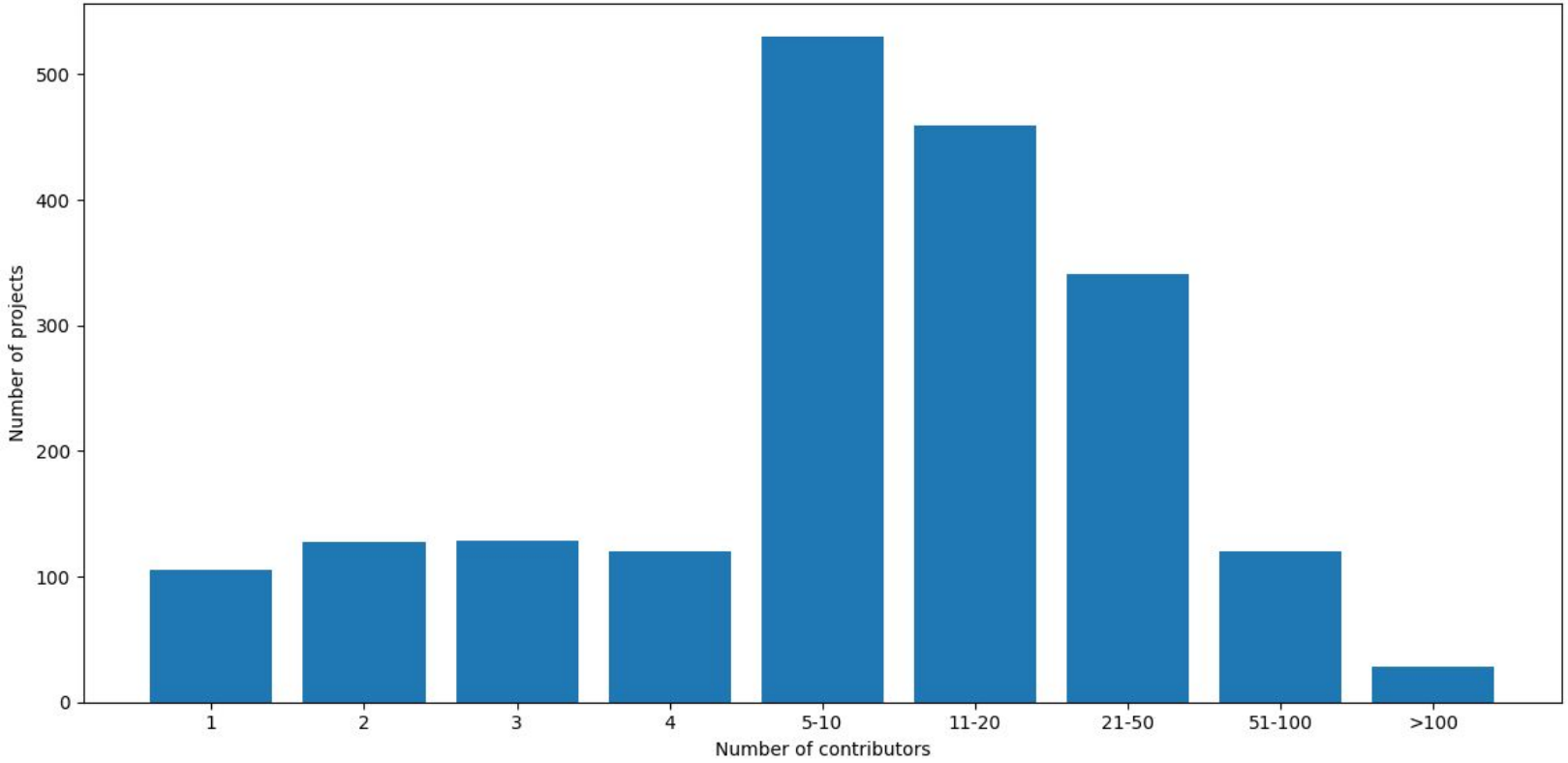
Basic statistics

- 2,497 total projects
- 2,372 on GitHub
- 70% active in the last year
- OST: 106 contributors, 84% by Tobias, 10~ other regular contributors

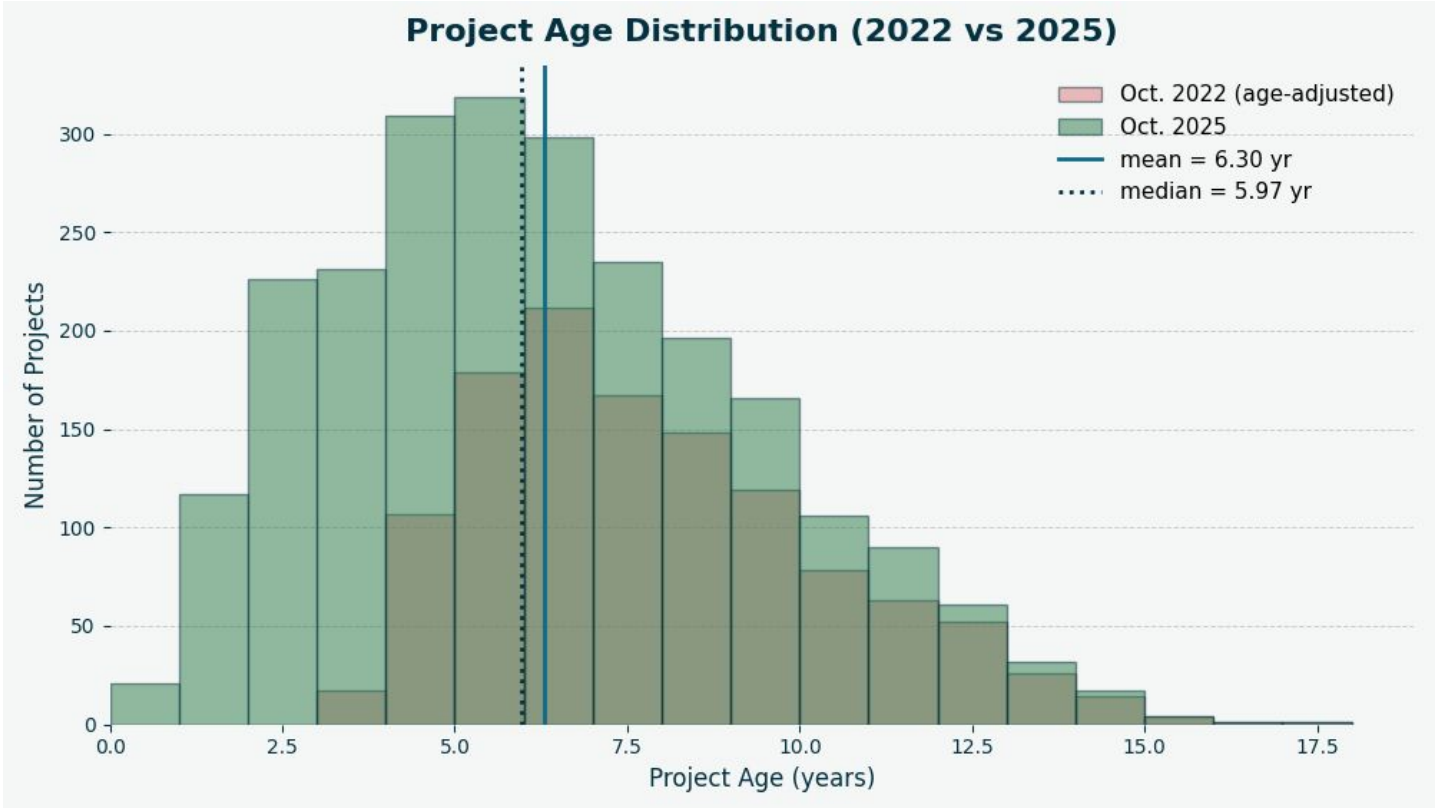
Programming languages



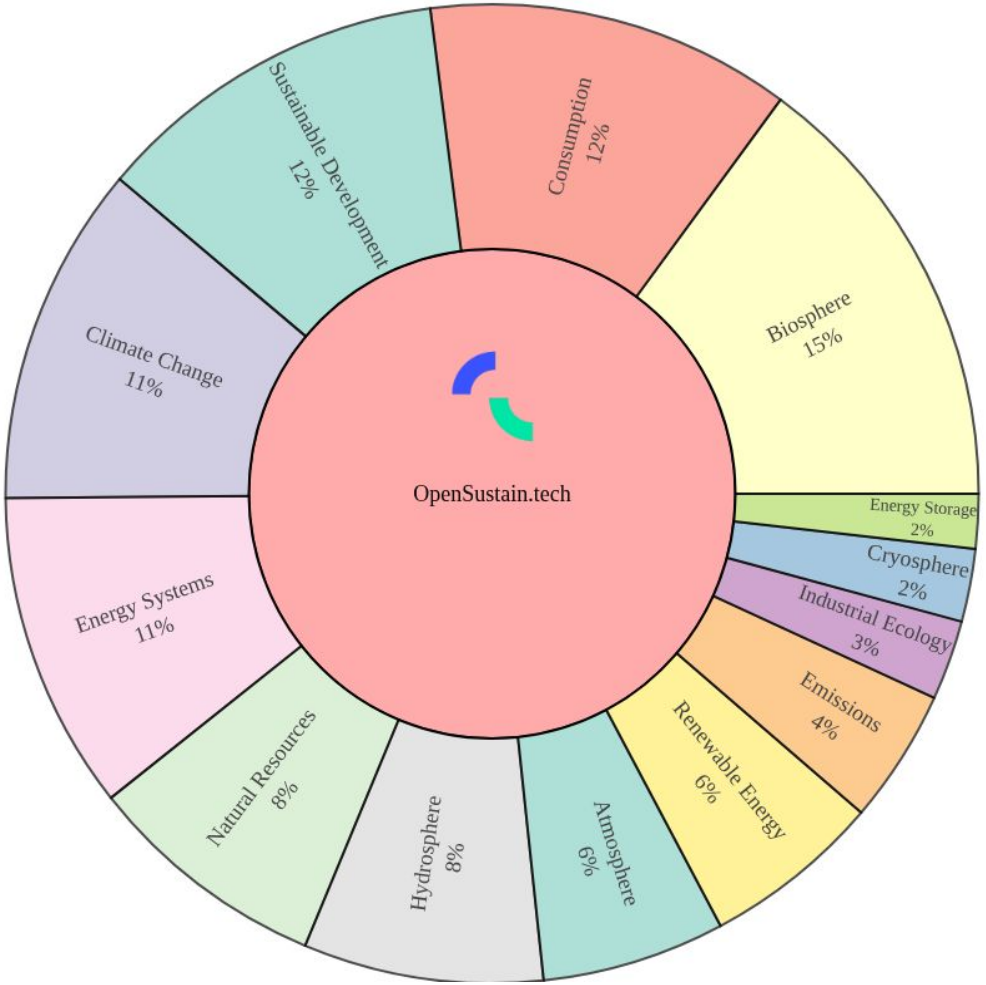
Number of contributors



Age distribution



Categories


OpenSustain.tech
Sustainable Development
12%
Consumption
12%
Biosphere
15%
Climate Change
11%
Energy Storage
2%
Cryosphere
2%
Industrial Ecology
3%
Emissions
4%
Renewable Energy
6%
Energy Systems
11%
Natural Resources
8%
Hydrosphere
8%
Atmosphere
6%

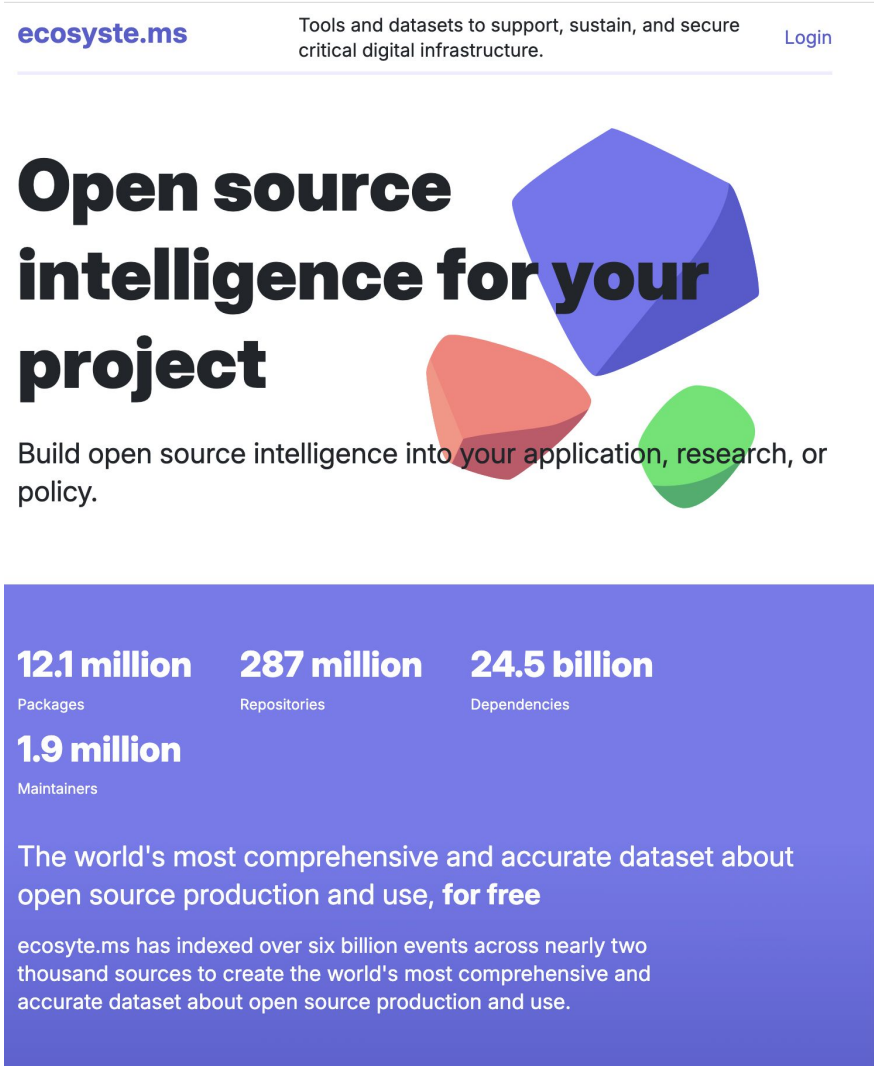
Dependencies

27,482 projects use OST-listed projects as deps

7,776 unique repositories; 2,715 unique packages

370 projects depend on other OST packages

52,559 dependencies; 14,518 unique packages



Ecosyste.ms Science Score

Average score is 43.07 (median 39).

1,829 (72.1%) of the projects had a Zenodo.json file

1,335 (52.6%) had academic emails in their commit messages;

959 (37.8%) had DOI references

859 (33.9%) had academic publication links.

343 (13.5%) had Citation.cff files

230 (9.1%) of the projects had an institutional organizational owner

172 (6.8%) of the projects were linked to JOSS papers.

Summary

OSS is definitely in use across the climate science ecosystem.

The economic argument for OSS needs to be fleshed out.

Better standards for citing and linking packages are necessary.

And we need more resourcing for OpenSustain.tech itself.

Thanks

Join us at https://opensustain.tech.

